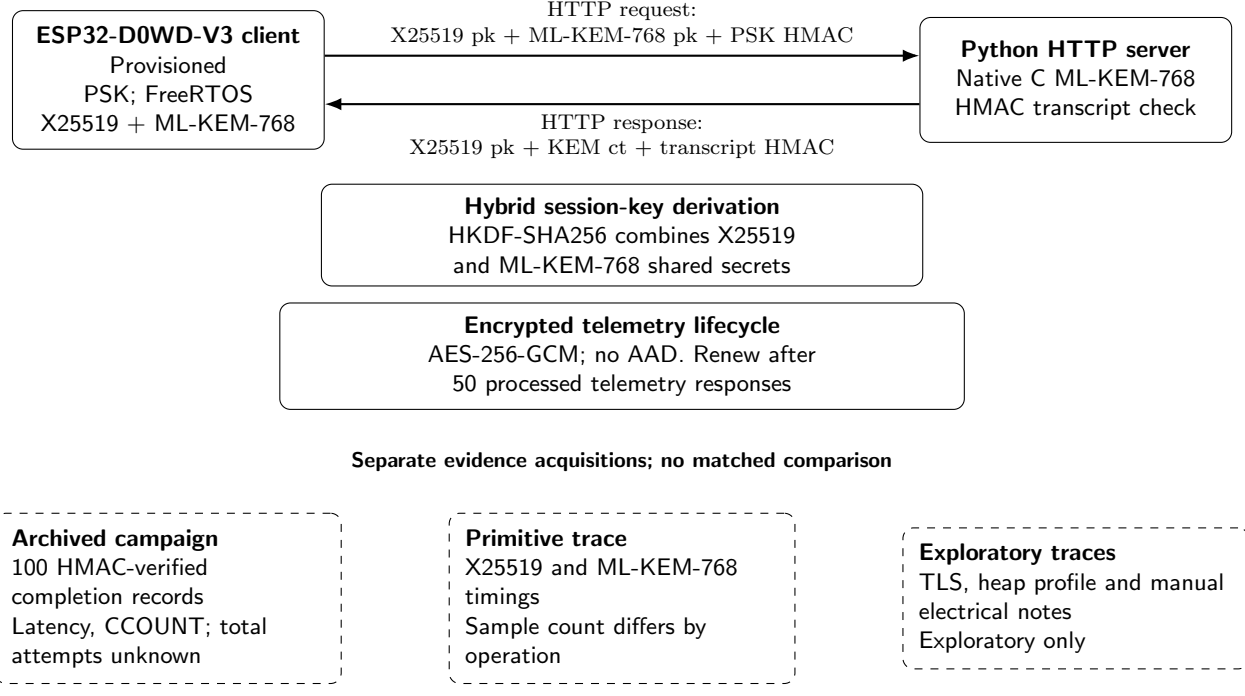
ESP32-D0WD-V3 client
Provisioned
PSK; FreeRTOS
X25519 + ML-KEM-768
HTTP request:
X25519 pk + ML-KEM-768 pk + PSK HMAC
HTTP response:
X25519 pk + KEM ct + transcript HMAC
Python HTTP server
Native C ML-KEM-768
HMAC transcript check
Hybrid session-key derivation
HKDF-SHA256 combines X25519
and ML-KEM-768 shared secrets
Encrypted telemetry lifecycle
AES-256-GCM; no AAD. Renew after
50 processed telemetry responses
Separate evidence acquisitions; no matched comparison
Archived campaign
100 HMAC-verified
completion records
Latency, CCOUNT; total
attempts unknown
Primitive trace
X25519 and ML-KEM-768
timings
Sample count differs by
operation
Exploratory traces
TLS, heap profile and manual
electrical notes
Exploratory only